P.1 Review of Real Numbers

Name _____ Period ____ Date _____

Pre-Calculus P.1 · Student sheet

I can…

- classify real numbers.
- write intervals in interval notation and as inequalities.
- use absolute value to find distance on the real number line.

Vocabulary · fill in the missing word

Rational number — a number that can be written as a ratio $\frac{p}{q}$ of _____, $q \neq 0$

Irrational number — a _____ number that is not rational

Interval notation — a way to write a set of real numbers with brackets and _____

Absolute value — a _____ distance from 0 on the real number line

Part 1 · Number sort

Numbers: -3, 0, $2/3$, $\sqrt{2}$, π, 5, -0.25, $\sqrt{9}$, $0.333\ldots$

Natural	Whole	Integer	Rational	Irrational

Part 2 · Notes and examples (fill in each note, then work the example)

Classifying: rational numbers have terminating or _____ decimals; irrational numbers do not.

Example 1

Which numbers in $\{-7, 0, \sqrt{5}, \frac{2}{3}, -1.6, \pi, 9, 0.333\ldots\}$ are (a) natural, (b) whole, (c) integers, (d) rational, (e) irrational?

Interval notation: a bracket [ ] includes an endpoint; a parenthesis ( ) _____ it. ∞ and $-\infty$ always get parentheses.

Example 2

Write each in interval notation.

a. $-2 < x \leq 3$

b. $x \geq 5$

c. $x < 0$

Example 3

Write an inequality and an interval: “x is at least 4 and less than 10.”

Absolute value: $|a| = a$ if $a \geq 0$, and $|a| =$ _____ if $a < 0$.

Example 4

Evaluate.

a. $|-3| + |-8|$

b. $-|-5|$

c. $|x|/x$ for $x < 0$

Distance between a and b on the number line: $d =$ _____

Example 5

Find the distance between -4 and 7 on the real number line.

Part 3 · Partner practice

Partner 1

Write $[-1, 6)$ as an inequality, and write $x > -3$ in interval notation.

Partner 2

Find the distance between -12 and -5.